Report On AI In Agriculture

1. Introduction – AI in Agriculture:

Artificial Intelligence (AI) is transforming the banking industry by making financial services Artificial Intelligence (AI) in agriculture refers to the use of machine learning, computer vision, robotics, and predictive analytics to improve farming practices. It enables farmers to make informed decisions, optimize resources, and increase productivity while reducing environmental impact.

2. How AI is Used in Agriculture:

Artificial intelligence (AI) is transforming agriculture by helping farmers make better decisions and improve crop production. AI can analyze weather patterns, monitor soil conditions, detect plant diseases, and identify pests at an early stage. It also supports smart irrigation and efficient use of fertilizers, reducing waste and increasing productivity. By using AI-powered tools, farmers can save time, lower costs, and produce healthier crops while promoting sustainable farming practices.

- **Satellite-based soil monitoring**: AI analyses hyperspectral satellite imagery to detect nutrient deficiencies instantly.
- **Precision drone spraying**: AI-powered drones identify weeds or pests and apply chemicals only where needed.
- **Generative AI assistants**: Tools like Krishi Mitra provide voice-based, localized advice to farmers.
- **AI crop insurance**: Systems like YES-TECH trigger automatic payouts based on satellite and drone data.
- **Weather forecasting**: AI predicts monsoon onset and helps farmers plan sowing.

3. Benefits of AI in Agriculture:

AI offers many benefits in agriculture by helping farmers make better decisions and improve productivity. It can monitor crop health, predict weather conditions, detect pests and diseases early, and optimize the use of water and fertilizers. AI also reduces labor costs, increases crop yields, and supports sustainable farming practices, making agriculture more efficient and environmentally friendly.

- **Higher productivity**: Precision farming reduces fertilizer use by up to 15% and pesticide use by 60% and also reduce human efforts.
- **Cost savings**: AI reduces unnecessary resource usage.
- **Risk management**: Automated crop insurance ensures timely payouts.
- **Accessibility**: Multilingual AI tools democratize expert knowledge for rural farmers.
- **Sustainability**: Reduced chemical use protects soil and water.

4. Challenges of AI in Agriculture:

AI in agriculture also faces several challenges. The high cost of AI technology and equipment makes it difficult for small farmers to adopt. Limited internet access, lack of technical knowledge, and concerns about data privacy can also slow its use. In addition, AI systems require accurate data and regular maintenance to work effectively, which can be challenging in rural areas.

- **Data heterogeneity**: Different soil, crop, and climate conditions make AI models hard to generalize.
- **Infrastructure gaps**: Rural areas may lack connectivity of internet for AI tools.
- **High costs**: Advanced drones and sensors are expensive for small farmers.
- **Trust and adoption**: Farmers may hesitate to rely on AI predictions.

5. Future Scope of AI in Agriculture:

The future of AI in agriculture is very promising. AI is expected to make farming smarter by improving crop production, reducing waste, and supporting precision farming. Future advancements may include fully automated farms, AI-powered robots, and better climate prediction systems. These innovations will help farmers increase efficiency, improve food security, and promote sustainable agriculture.

- **Multimodal data fusion**: Combining satellite, sensor, and drone data for holistic insights.
- **Cloud-edge collaboration**: Real-time processing at the farm level.
- **Robotic harvesting**: Autonomous robots for fruit and vegetable picking.
- **AI-driven marketplaces**: Linking farmers directly to buyers with predictive pricing.

6. Latest Technologies Used in AI in Agriculture:

- **Bharat-VISTAAR**: A multilingual AI-driven advisory portal integrating AgriStack and ICAR research.
  BharatVistaar

- **Krishi Mitra** : A generative AI assistant for farmers, available in local dialects.
  KrishiMitra — AI Farming Companion

- **Precision Drone Pollination (BrijBot)**: AI robots for targeted spraying and pollination.

- **YES-TECH Crop Insurance**: AI-based yield estimation system for automatic payouts.
  YES-TECH

- **OneSoil** : Satellite Soil Health Monitoring - Instant nutrient analysis through hyperspectral imaging.
  OneSoil — Satellite Agricultural Intelligence | OneSoil

- **Kisaantrade.com :** An online platform that connects all the agriculture companies, new start-up companies and agri business owners of India on one platform.
  Kisaan Trade | Global Agriculture B2B Online Marketplace

7. References:

- **MDPI Journal** – *Artificial Intelligence in Smart Agriculture Across the Production-to-Postharvest Continuum* (2026).

  Artificial Intelligence in Smart Agriculture Across the Production-to-Postharvest Continuum: Progress, Challenges, and Future Directions

- Top 10 AI Innovations in Agriculture & Rural Livelihoods (2026).

  Top 10 AI Innovations in Agriculture & Rural Livelihoods -

- AI Transforming Indian Agriculture – Digital Agriculture Mission (2026).

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